

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q12-0173

Ranch 138 kV
April 28, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0173. The Interconnection Customer proposes a 389.5 MW natural gas combined cycle facility interconnecting at the Ranch 138 kV substation in the Pennsylvania area. The request was received on November 15, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 125 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,278,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The WILIS-RANCH 138 KV L is overloaded to 160.3% of its 238 MVA emergency rating for the loss of the RANCH-TRI-CORNER SUBSTATION 138 KV. The project contributes 100.0% DFAX (389.5 MW) to this overload. RANCH-TRI-CORNER 138 KV: 152.0% loading (251 MVA) following the outage of the WILIS-RANCH 138 KV L; DFAX = 100.0%.

For the loss of WILIS-RANCH 138 KV, the WOODLANDS-TRI-CORNER 138 KV Line loads to 119.4% of its 250 MVA rating. Project DFAX: 68.9%.

For the loss of E DEVON SUBSTATION-THIELE KAOLIN REEDY CRK 345 KV, the SPINDLE 500/345kV TX loads to 106.9% of its 750 MVA rating. Project DFAX: 5.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RANCH SUBSTATION-WILIS 138 KV	Rebuild line with higher-capacity conductor	\$5,431,000	100% (\$5,431,000)
RANCH-TRI-CORNER 138 KV	Rebuild line with higher-capacity conductor	\$1,452,000	100% (\$1,452,000)
WOODLANDS-TRI-CORNER 138 KV	Rebuild line with higher-capacity conductor	\$3,988,000	100% (\$3,988,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SPINDLE 500/345kV TX	Replace transformer with higher-rated unit	\$36,335,000	12% (\$4,407,000)
	Total allocated to project		\$15,278,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.